

1. Problem 1

- **Predict**- Which of the distributions from Week 1 could plausibly have produced this sample, and which are ruled out? Name the moment that rules out each one you reject.

In problem 1, the mean is 0.00110439, var is 0.00009625, skew is -0.6698, kurtosis is 2.3576.

The normal distribution, lognormal distribution and student's t distribution are ruled out.

In normal distribution, the skewness and excess kurtosis need to be 0

In lognormal distribution, the skewness needs to be positive

In t distribution, the skewness needs to be 0

Only NIG is still plausible

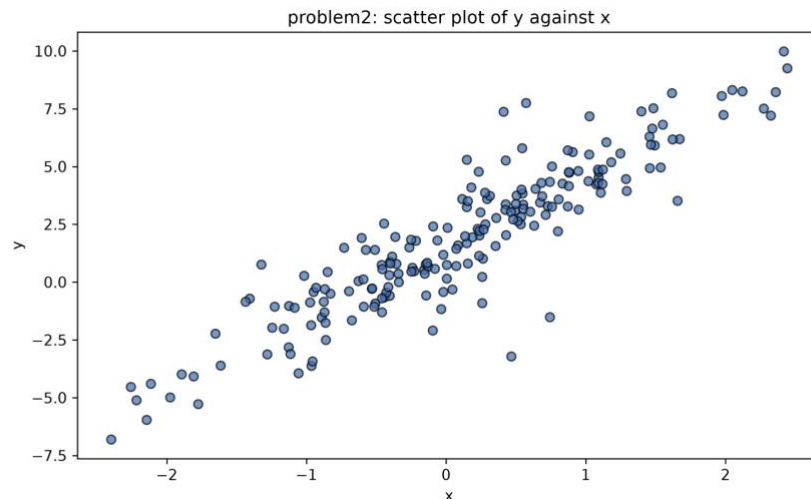
- **Fit**- How many observations fall below the Normal's 1% quantile? How many should there be? Report both numbers.

There are 26 observations in real appearing. 10 should be there in normal under 1%.

- **Reconcile** – If we used this fitted Normal as our model of the series, what would it get wrong? Say where in the distribution the error shows up, and in which direction.

The fitted normal matches the sample mean and variance by construction. However, 26 observations fall below estimated 1% quantile, compared with 10 under the normal. Therefore, the normal model underestimates the left tail risk.

2. Problem 2



- a. Describe what you see in the scatter. What distribution do you expect the error term to follow, and why?

The scatter suggests a positive and approximately linear relationship. Most observations appear close to the central trend, but there are several unusually large vertical deviation. I think the errors are to follow a heavy tail student's t distribution

- b. Ordinary least squares assumes normally distributed errors. Name which of the seven OLS assumptions this data violates, and predict what that violation will do – to the estimated slope, and separately to the standard error of that slope.

This data violates OLS assumption 7, which states the error term is normally distributed. I think the heavy tail errors have little effect on the estimated slope because of clear linear relationship. However, the heavy-tailed errors can increase the slope's standard error and make Normal-based inference less reliable.

- c. Compare the three estimates of β . Did the violation move the slope as much as you predicted?

The OLS and Normal MLE estimates of beta are both approximately 2.99, while the student's t estimate is around 3.019. therefore, the estimated slopes are very close. This agrees with my prediction.

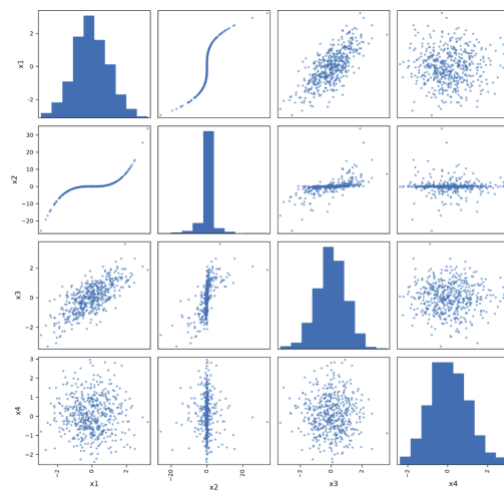
- d. The AICc prefers one model by a wide margin, yet the slopes are close together. What, specifically, is the rejected model getting wrong, if not the slope?

The student t model has a substantially lower AIC than the normal distribution. The gap is around 26. Although both models estimate nearly the same slope, the normal model assigns too little probability to large residuals.

- e. Compare the 95% and the 99.5% quantiles of the fitted Normal error against those of the fitted t error. One is wider at 95% and the other is wider at 99.5%. Explain why, and say which one you would want if you were setting a capital buffer.

The Normal has a wider 95% quantile, while the student's t has a wider 99.5% quantile because the distribution has a narrower center but heavier tails. I would use the student's t for a capital buffer because it better captures extreme losses.

3. Problem 3



- a. Which pair do you expect the Pearson and Spearman coefficients to disagree on most? Which pairs do you expect them to agree on? Say why in each case.

I expect x_1 and x_2 have the largest disagreement. Their relationship is strongly monotonic but non linear, so its Spearman should be higher than Person. And I think the coefficients to agree more for x_1 and x_3 , which have approximately linear relationship.

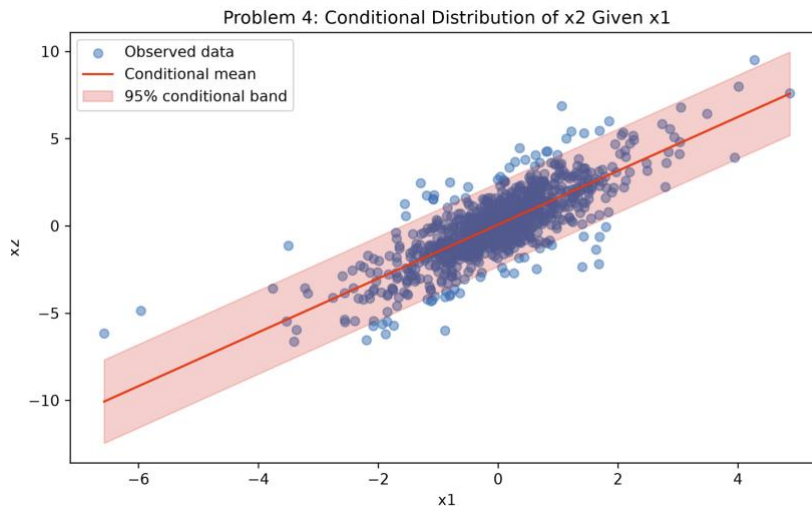
- b. Compute both correlation matrices. Report the pair with the largest gap.

The largest gap is between x_1 and x_2 . Their Pearson correlation is 0.799, while their Spearman cor is 0.9719, giving a gap of 0.1728

- c. Explain the mechanism that produces the gap. Which of the two coefficients is the honest description of that pair, and what question is the other one answering?

The gap occurs because x_1 and x_2 have a strong monotonic but nonlinear relationship. Spearman is the more honest description because it captures their consistent rank ordering.

4. Problem 4



- a. Write down the conditional variance of x_2 given x_1 using the partitioned result from Week 2, in the blocks of Σ . Say which of your series is which block. If you learn the value of x_1 , by what factor does your uncertainty about x_2 fall? Give that factor as a formula in the same blocks, and then as a number for this data.

x_1 corresponds to block 1 and x_2 corresponds to block 2. The conditional variance is using the sample covariance matrix, the conditional variance is 1.4867. Relative to the original variance of x_2 , the remaining variance factor is 0.3622.

Therefore, observing x_1 reduces the variance of x_2 by approximately 63.78%.

- b. Under the multivariate normal, does that factor depend on the value of x_1 you observed? Point at the term in the formula that settles it, and answer before you look at the data.

Under the multivariate normal assumption, this factor does not depend on the observed value of x_1 . The conditional variance formula contains only the covariance blocks and contains no x_1 term.

- c. Write down the conditional mean of x_2 given x_1 , in the same blocks of Σ . Identify the coefficient on x_1 and say what it is in regression terms. Then plot the conditional expectation across the range of x_1 , with a 95% band around it, over a scatter of the data.

The conditional mean is $\mu_2 + \Sigma_{21} \Sigma_{11}^{-1} (x_1 - \mu_1)$. For this data, the coefficient on (x_1) is 1.5428.

This coefficient is the slope from regressing (x_2) on (x_1) . The conditional mean and its constant-width 95% band are shown in the plot.

- d. Compute the fraction of observations that fall inside the band. Report it.

Overall, 93.50% of observations fall inside the 95% conditional band. This is slightly below the theoretical 95% coverage.

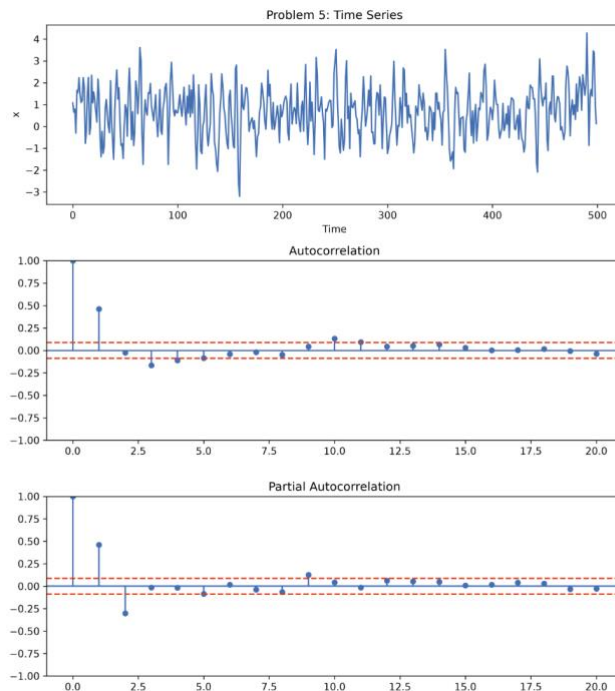
- e. Now split the sample by how far x_1 sits from its mean – inside one standard deviation, between one and two, and beyond two – and compute the coverage within each bucket. Report all three

Coverage within one standard deviation is 95.65%. Coverage between one and two standard deviations is 85.79%, and coverage beyond two standard deviations is 90.20%.

- f. The coverage is not flat across those buckets. Explain what that tells you about the data, and identify which assumption behind the constant width band has failed. Note carefully which parts of your answer to part (c) survive the failure and which do not.

Coverage is not constant across the three buckets, especially between one and two standard deviations. This suggests that the conditional error distribution changes with x_1 . Therefore, the constant conditional variance assumption implied by the multivariate Normal model has failed. The fitted linear regression relationship and slope may still be useful, but the constant-width band and its theoretical 95% coverage are no longer reliable.

5. Problem 5



- Which of the two functions cuts off, and which decays? What order does that imply, and is it an AR or an MA process? Commit to a specific answer before fitting.
- State the rule you used. Cite the significance band you judged “cuts off” against.

The PACF cuts off after lag 2, while the ACF gradually decays. Therefore, I predict an AR(2) process. I used the rule that an AR(p) process has a PACF that cuts off after lag (p), while its ACF decays. I judged significance using the approximate 95% band.

- Fit AR(1), AR(2), AR(3), MA(1), MA(2), and MA(3). Report the AICc for each.

Model	AICc
AR(1)	1418.5870
AR(2)	1372.4383
AR(3)	1374.3432
MA(1)	1389.0843
MA(2)	1381.2316
MA(3)	1374.1165

- Which model does the AICc select? Did it match your prediction?

AICc selects AR(2), which matches my prediction from the ACF and PACF plots.

- e. Compare the AR(2) and AR(3) fits. The third coefficient is small. Explain what AICc is trading off when it rejects the larger model, and why R^2 would not have made the same choice.

In the AR(3) model, the third coefficient is only (-0.0165), so it provides little improvement in fit. AICc penalizes the additional parameter and therefore prefers AR(2).

In contrast, (R^2) does not penalize model complexity and cannot decrease when another predictor is added, so it would not favor the simpler model for this reason.